

## SAFETY DATA SHEET

Revision Date 25-March-2024

Revision Number 2

### 1. Identification

**Product Name** 1-Bromo-2-butene

**Cat No. :** R21762

**CAS-No** 4784-77-4  
**Synonyms** No information available

**Recommended Use** Laboratory chemicals.  
**Uses advised against** Food, drug, pesticide or biocidal product use.

#### Details of the supplier of the safety data sheet

##### Company

**Importer/Distributor**  
Fisher Scientific  
112 Colonnade Road,  
Ottawa, ON K2E 7L6,  
Canada  
Tel: 1-800-234-7437

##### **Emergency Telephone Number**

For information **US** call: 001-800-227-6701 / **Europe** call: +32 14 57 52 11  
Emergency Number **US**:001-201-796-7100 / **Europe**: +32 14 57 52 99  
**CHEMTREC** Tel. No. **US**:001-800-424-9300 / **Europe**:001-703-527-3887

### 2. Hazard(s) identification

#### Classification

**WHMIS 2015 Classification** Classified as hazardous under the Hazardous Products Regulations (SOR/2015-17)

|                                          |            |
|------------------------------------------|------------|
| <b>Flammable liquids</b>                 | Category 2 |
| <b>Skin Corrosion/Irritation</b>         | Category 2 |
| <b>Serious Eye Damage/Eye Irritation</b> | Category 2 |

#### Label Elements

##### **Signal Word**

Danger

##### **Hazard Statements**

Highly flammable liquid and vapor  
Causes skin irritation  
Causes serious eye irritation

**Precautionary Statements****Prevention**

Keep away from heat, hot surfaces, sparks, open flames and other ignition sources. No smoking

Keep container tightly closed

Ground/bond container and receiving equipment

Wash face, hands and any exposed skin thoroughly after handling

Wear protective gloves/protective clothing/eye protection/face protection

Use non-sparking tools

Take action to prevent static discharges

**Response**

IF ON SKIN (or hair): Take off immediately all contaminated clothing. Rinse skin with water/ shower

IF IN EYES: Rinse cautiously with water for several minutes. Remove contact lenses, if present and easy to do. Continue rinsing

If skin irritation occurs: Get medical advice/attention

If eye irritation persists: Get medical advice/attention

In case of fire: Use dry sand, dry chemical or alcohol-resistant foam to extinguish

Take off contaminated clothing and wash it before reuse

**Storage**

Store in a well-ventilated place. Keep cool

**Disposal**

Dispose of contents/container to an approved waste disposal plant

### 3. Composition/Information on Ingredients

| Component          | CAS-No    | Weight % |
|--------------------|-----------|----------|
| 2-Butene, 1-bromo- | 4784-77-4 | <=100    |

### 4. First-aid measures

|                                        |                                                                                                                                                    |
|----------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>General Advice</b>                  | If symptoms persist, call a physician.                                                                                                             |
| <b>Eye Contact</b>                     | Rinse immediately with plenty of water, also under the eyelids, for at least 15 minutes. Get medical attention.                                    |
| <b>Skin Contact</b>                    | Wash off immediately with plenty of water for at least 15 minutes. If skin irritation persists, call a physician.                                  |
| <b>Inhalation</b>                      | Remove to fresh air. If not breathing, give artificial respiration. Get medical attention if symptoms occur.                                       |
| <b>Ingestion</b>                       | Clean mouth with water and drink afterwards plenty of water.                                                                                       |
| <b>Most important symptoms/effects</b> | None reasonably foreseeable. . Inhalation of high vapor concentrations may cause symptoms like headache, dizziness, tiredness, nausea and vomiting |
| <b>Notes to Physician</b>              | Treat symptomatically                                                                                                                              |

### 5. Fire-fighting measures

|                                     |                                                   |
|-------------------------------------|---------------------------------------------------|
| <b>Suitable Extinguishing Media</b> | Water mist may be used to cool closed containers. |
|-------------------------------------|---------------------------------------------------|

|                                         |                          |
|-----------------------------------------|--------------------------|
| <b>Unsuitable Extinguishing Media</b>   | No information available |
| <b>Flash Point</b>                      | 11 °C / 51.8 °F          |
| <b>Method -</b>                         | No information available |
| <b>Autoignition Temperature</b>         | No information available |
| <b>Explosion Limits</b>                 |                          |
| <b>Upper</b>                            | No data available        |
| <b>Lower</b>                            | No data available        |
| <b>Sensitivity to Mechanical Impact</b> | No information available |
| <b>Sensitivity to Static Discharge</b>  | No information available |

**Specific Hazards Arising from the Chemical**

Flammable. Containers may explode when heated. Vapors may form explosive mixtures with air. Vapors may travel to source of ignition and flash back.

**Hazardous Combustion Products**

Carbon oxides. Hydrogen bromide.

**Protective Equipment and Precautions for Firefighters**

As in any fire, wear self-contained breathing apparatus pressure-demand, MSHA/NIOSH (approved or equivalent) and full protective gear.

**NFPA**

**Health**  
2

**Flammability**  
3

**Instability**  
0

**Physical hazards**  
N/A

## 6. Accidental release measures

|                                             |                                                                                                                                                                               |
|---------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Personal Precautions</b>                 | Ensure adequate ventilation. Use personal protective equipment as required. Remove all sources of ignition. Take precautionary measures against static discharges.            |
| <b>Environmental Precautions</b>            | Should not be released into the environment.                                                                                                                                  |
| <b>Methods for Containment and Clean Up</b> | Soak up with inert absorbent material. Keep in suitable, closed containers for disposal. Remove all sources of ignition. Use spark-proof tools and explosion-proof equipment. |

## 7. Handling and storage

|                 |                                                                                                                                                                                                                                                                                                                                                                                                                                     |
|-----------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Handling</b> | Wear personal protective equipment/face protection. Ensure adequate ventilation. Do not get in eyes, on skin, or on clothing. Avoid ingestion and inhalation. Keep away from open flames, hot surfaces and sources of ignition. Use only non-sparking tools. To avoid ignition of vapors by static electricity discharge, all metal parts of the equipment must be grounded. Take precautionary measures against static discharges. |
| <b>Storage.</b> | Keep container tightly closed in a dry and well-ventilated place. Keep away from heat, sparks and flame.                                                                                                                                                                                                                                                                                                                            |

## 8. Exposure controls / personal protection

|                             |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
|-----------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Exposure Guidelines</b>  | This product does not contain any hazardous materials with occupational exposure limits established by the region specific regulatory bodies.                                                                                                                                                                                                                                                                                                                                                                         |
| <b>Engineering Measures</b> | Ensure that eyewash stations and safety showers are close to the workstation location. Ensure adequate ventilation, especially in confined areas. Use explosion-proof electrical/ventilating/lighting/equipment. Wherever possible, engineering control measures such as the isolation or enclosure of the process, the introduction of process or equipment changes to minimise release or contact, and the use of properly designed ventilation systems, should be adopted to control hazardous materials at source |

**Personal protective equipment****Eye Protection**  
**Hand Protection**Goggles  
Wear appropriate protective gloves and clothing to prevent skin exposure.

| Glove material | Breakthrough time | Glove thickness | Glove comments         |
|----------------|-------------------|-----------------|------------------------|
| Nitrile rubber | See manufacturers | -               | Splash protection only |
| Neoprene       | recommendations   |                 |                        |
| Natural rubber |                   |                 |                        |
| PVC            |                   |                 |                        |

Inspect gloves before use. observe the instructions regarding permeability and breakthrough time which are provided by the supplier of the gloves. (Refer to manufacturer/supplier for information) gloves are suitable for the task: Chemical compatability, Dexterity, Operational conditions, User susceptibility, e.g. sensitisation effects, also take into consideration the specific local conditions under which the product is used, such as the danger of cuts, abrasion. gloves with care avoiding skin contamination.

**Respiratory Protection**

When workers are facing concentrations above the exposure limit they must use appropriate certified respirators. Follow the OSHA respirator regulations found in 29 CFR 1910.134 or European Standard EN 149. Use a NIOSH/MSHA or European Standard EN 149 approved respirator if exposure limits are exceeded or if irritation or other symptoms are experienced.

To protect the wearer, respiratory protective equipment must be the correct fit and be used and maintained properly

**Recommended Filter type:** Organic gases and vapours filter Type A Brown conforming to EN14387

When RPE is used a face piece Fit Test should be conducted

**Environmental exposure controls**

No information available.

**Hygiene Measures**

Handle in accordance with good industrial hygiene and safety practice. Keep away from food, drink and animal feeding stuffs. Do not eat, drink or smoke when using this product. Remove and wash contaminated clothing and gloves, including the inside, before re-use. Wash hands before breaks and after work.

## 9. Physical and chemical properties

|                                        |                               |
|----------------------------------------|-------------------------------|
| Physical State                         | Liquid                        |
| Appearance                             | No information available      |
| Odor                                   | No information available      |
| Odor Threshold                         | No information available      |
| pH                                     | Not applicable                |
| Melting Point/Range                    | No data available             |
| Boiling Point/Range                    | 105 - 107 °C / 221 - 224.6 °F |
| Flash Point                            | 11 °C / 51.8 °F               |
| Evaporation Rate                       | No information available      |
| Flammability (solid,gas)               | Not applicable                |
| Flammability or explosive limits       |                               |
| Upper                                  | No data available             |
| Lower                                  | No data available             |
| Vapor Pressure                         | No information available      |
| Vapor Density                          | No information available      |
| Density                                | 1.31                          |
| Specific Gravity                       | No information available      |
| Solubility                             | No information available      |
| Partition coefficient; n-octanol/water | No data available             |
| Autoignition Temperature               | No information available      |
| Decomposition Temperature              | No information available      |
| Viscosity                              | No information available      |
| Molecular Formula                      | C4 H7 Br                      |
| Molecular Weight                       | 135.00                        |

## 10. Stability and reactivity

|                                         |                                                                   |
|-----------------------------------------|-------------------------------------------------------------------|
| <b>Reactive Hazard</b>                  | None known, based on information available                        |
| <b>Stability</b>                        | Stable under normal conditions.                                   |
| <b>Conditions to Avoid</b>              | Keep away from open flames, hot surfaces and sources of ignition. |
| <b>Incompatible Materials</b>           | Strong oxidizing agents                                           |
| <b>Hazardous Decomposition Products</b> | Carbon oxides, Hydrogen bromide                                   |
| <b>Hazardous Polymerization</b>         | Hazardous polymerization does not occur.                          |
| <b>Hazardous Reactions</b>              | None under normal processing.                                     |

## 11. Toxicological information

### Acute Toxicity

#### Product Information

#### Component Information

|                                             |                          |
|---------------------------------------------|--------------------------|
| <b>Toxicologically Synergistic Products</b> | No information available |
|---------------------------------------------|--------------------------|

#### Delayed and immediate effects as well as chronic effects from short and long-term exposure

|                        |                                                                                          |
|------------------------|------------------------------------------------------------------------------------------|
| <b>Irritation</b>      | No information available                                                                 |
| <b>Sensitization</b>   | No information available                                                                 |
| <b>Carcinogenicity</b> | The table below indicates whether each agency has listed any ingredient as a carcinogen. |

| Component          | CAS-No    | IARC       | NTP        | ACGIH      | OSHA       | Mexico     |
|--------------------|-----------|------------|------------|------------|------------|------------|
| 2-Butene, 1-bromo- | 4784-77-4 | Not listed | Not listed | Not listed | Not listed | Not listed |

|                          |                          |
|--------------------------|--------------------------|
| <b>Mutagenic Effects</b> | No information available |
|--------------------------|--------------------------|

|                             |                           |
|-----------------------------|---------------------------|
| <b>Reproductive Effects</b> | No information available. |
|-----------------------------|---------------------------|

|                              |                           |
|------------------------------|---------------------------|
| <b>Developmental Effects</b> | No information available. |
|------------------------------|---------------------------|

|                       |                           |
|-----------------------|---------------------------|
| <b>Teratogenicity</b> | No information available. |
|-----------------------|---------------------------|

|                               |            |
|-------------------------------|------------|
| <b>STOT - single exposure</b> | None known |
|-------------------------------|------------|

|                                 |            |
|---------------------------------|------------|
| <b>STOT - repeated exposure</b> | None known |
|---------------------------------|------------|

|                          |                          |
|--------------------------|--------------------------|
| <b>Aspiration hazard</b> | No information available |
|--------------------------|--------------------------|

|                                                   |                                                                                                                     |
|---------------------------------------------------|---------------------------------------------------------------------------------------------------------------------|
| <b>Symptoms / effects, both acute and delayed</b> | Inhalation of high vapor concentrations may cause symptoms like headache, dizziness, tiredness, nausea and vomiting |
|---------------------------------------------------|---------------------------------------------------------------------------------------------------------------------|

|                                        |                          |
|----------------------------------------|--------------------------|
| <b>Endocrine Disruptor Information</b> | No information available |
|----------------------------------------|--------------------------|

|                              |                                                                |
|------------------------------|----------------------------------------------------------------|
| <b>Other Adverse Effects</b> | The toxicological properties have not been fully investigated. |
|------------------------------|----------------------------------------------------------------|

## 12. Ecological information

### Ecotoxicity

Do not empty into drains.

|                                      |                          |
|--------------------------------------|--------------------------|
| <b>Persistence and Degradability</b> | No information available |
|--------------------------------------|--------------------------|

**Bioaccumulation/ Accumulation** No information available.

**Mobility** No information available.

### 13. Disposal considerations

**Waste Disposal Methods** Chemical waste generators must determine whether a discarded chemical is classified as a hazardous waste. Chemical waste generators must also consult local, regional, and national hazardous waste regulations to ensure complete and accurate classification.

### 14. Transport information

#### DOT

UN-No UN1993  
 Proper Shipping Name FLAMMABLE LIQUIDS, N.O.S.  
 Hazard Class 3  
 Packing Group II

#### TDG

UN-No UN1993  
 Proper Shipping Name FLAMMABLE LIQUID, N.O.S.  
 Hazard Class 3  
 Packing Group II

#### IATA

UN-No UN1993  
 Proper Shipping Name FLAMMABLE LIQUID, N.O.S.  
 Hazard Class 3  
 Packing Group II

#### IMDG/IMO

UN-No UN1993  
 Proper Shipping Name FLAMMABLE LIQUID, N.O.S.  
 Hazard Class 3  
 Packing Group II

### 15. Regulatory information

All of the components in the product are on the following Inventory lists: X = listed U.S.A. (TSCA) Canada (DSL/NDSL) Europe (EINECS/ELINCS/NLP) Australia (AICS) Korea (KECL) China (IECSC) Japan (ENCS) Philippines (PICCS)

#### International Inventories

| Component          | CAS-No    | DSL | NDSL | TSCA | TSCA Inventory notification - Active-Inactive | EINECS    | ELINCS | NLP |
|--------------------|-----------|-----|------|------|-----------------------------------------------|-----------|--------|-----|
| 2-Butene, 1-bromo- | 4784-77-4 | -   | X    | X    | INACTIVE                                      | 225-332-5 | -      | -   |

| Component          | CAS-No    | IECSC | KECL | ENCS | ISHL | TCSI | AICS | NZIoC | PICCS |
|--------------------|-----------|-------|------|------|------|------|------|-------|-------|
| 2-Butene, 1-bromo- | 4784-77-4 | -     | -    | -    | -    | X    | -    | -     | -     |

#### Legend:

X - Listed '-' - Not Listed

KECL - NIER number or KE number (<http://ncis.nier.go.kr/en/main.do>)

DSL/NDSL - Canadian Domestic Substances List/Non-Domestic Substances List

TSCA - United States Toxic Substances Control Act Section 8(b) Inventory

EINECS/ELINCS - European Inventory of Existing Commercial Chemical Substances/EU List of Notified Chemical Substances

IECSC - Chinese Inventory of Existing Chemical Substances

KECL - Korean Existing and Evaluated Chemical Substances

ENCS - Japanese Existing and New Chemical Substances

AICS - Australian Inventory of Chemical Substances

PICCS - Philippines Inventory of Chemicals and Chemical Substances

#### Canada

SDS in compliance with provisions of information as set out in Canadian Standard - Part 4, Schedule 1 and 2 of the Hazardous Products Regulations (HPR) and meets the requirements of the HPR (Paragraph 13(1)(a) of the Hazardous Products Act (HPA)).

#### Other International Regulations

Authorisation/Restrictions according to EU REACH Not applicable

#### Safety, health and environmental regulations/legislation specific for the substance or mixture

| Component          | CAS-No    | OECD HPV       | Persistent Organic Pollutant | Ozone Depletion Potential | Restriction of Hazardous Substances (RoHS) |
|--------------------|-----------|----------------|------------------------------|---------------------------|--------------------------------------------|
| 2-Butene, 1-bromo- | 4784-77-4 | Not applicable | Not applicable               | Not applicable            | Not applicable                             |

| Component          | CAS-No    | Seveso III Directive (2012/18/EC) - Qualifying Quantities for Major Accident Notification | Seveso III Directive (2012/18/EC) - Qualifying Quantities for Safety Report Requirements | Rotterdam Convention (PIC) | Basel Convention (Hazardous Waste) |
|--------------------|-----------|-------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------|----------------------------|------------------------------------|
| 2-Butene, 1-bromo- | 4784-77-4 | Not applicable                                                                            | Not applicable                                                                           | Not applicable             | Not applicable                     |

### 16. Other information

**Prepared By** Product Safety Department  
Email: chem.techinfo@thermofisher.com  
www.thermofisher.com

**Revision Date** 25-March-2024  
**Print Date** 25-March-2024  
**Revision Summary** New emergency telephone response service provider.

#### Disclaimer

The information provided in this Safety Data Sheet is correct to the best of our knowledge, information and belief at the date of its publication. The information given is designed only as a guidance for safe handling, use, processing, storage, transportation, disposal and release and is not to be considered a warranty or quality specification. The information relates only to the specific material designated and may not be valid for such material used in combination with any other materials or in any process, unless specified in the text

**End of SDS**